⚡ Section: Electricity

🔷 Topic: Internal Resistance

Given:

A battery of EMF $E = 12.0\,\mathrm{V}$ and internal resistance r is connected in a circuit with three $16\,\Omega$ resistors in **parallel**.
Ammeter reading: $I = 0.38\,\mathrm{A}$

(a)(i) Determine the terminal potential difference (t.p.d.)

✔ 2 marks

First, calculate total external resistance:

$$\frac{1}{R_{\text{ext}}} = \frac{1}{16} + \frac{1}{16} + \frac{1}{16} = \frac{3}{16} \quad \Rightarrow \quad R_{\text{ext}} = \frac{16}{3} = 5.33\,\Omega$$

Then use:
$V = IR_{\text{ext}} = 0.38 \times 5.33 = 2.03\,\mathrm{V}$

(a)(ii) Calculate the internal resistance of the battery

✔ 3 marks

Use:
$E = V + Ir \quad \Rightarrow \quad 12 = 2.03 + 0.38r$

$$r = \frac{12 - 2.03}{0.38} = \frac{9.97}{0.38} = 26.2\,\Omega$$

(a)(iii) Calculate the power dissipated by the internal resistance

✔ 3 marks

$$P = I^2 r = (0.38)^2 \times 26.2 = 0.1444 \times 26.2 = 3.79\,\mathrm{W}$$

(b) Compare power dissipated if circuit is rearranged in series

✔ 2 marks

New configuration: Resistors in **series**, total external resistance:
$R = 16 + 16 + 16 = 48\,\Omega$

New total resistance:
$R_{\text{total}} = 48 + 26.2 = 74.2\,\Omega$

New current:
$$I = \frac{E}{R_{\text{total}}} = \frac{12}{74.2} = 0.162\,\mathrm{A}$$

New power:
$P = I^2 r = (0.162)^2 \times 26.2 = 0.0262 \times 26.2 = 0.687\,\mathrm{W}$

✅ Answer:

> The power dissipated by the internal resistance is **less**, since the current in the circuit is lower when the resistors are in series, and power loss depends on $I^2 r$.

🧠 Revision Tips

- Use parallel resistance rules carefully: $\frac{1}{R} = \sum \frac{1}{R_i}$
- Power loss in batteries occurs due to internal resistance: $P = I^2 r$
- The internal resistance becomes more significant when current is high (parallel circuits)
- Terminal p.d. is always less than EMF when current flows: $V = E - Ir$